Name_____ Date_____ Period_____

LESSON 6-7 · HOMEWORK PACKET · SHEET 1 OF 2

Factors and Greatest Common Factor

Maryland Standard 6.NS.B.4 · Learning target – I can list the factors of two numbers and find their greatest common factor.

REMEMBER

- A **factor** divides a number with no remainder. Find factors in pairs: 18 = 1 × 18 = 2 × 9 = 3 × 6.
- The **greatest common factor** (GCF) is the greatest factor that two numbers share.
- Use the GCF to make the **greatest number of identical groups** with nothing left over.

WORKED EXAMPLE

Find the GCF of 24 and 36.

Step 1 Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24

Step 2 Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36

Step 3 Common factors: 1, 2, 3, 4, 6, 12. The greatest is 12.

Answer **GCF = 12**

PRACTICE

show your work · label units

1. List all the factors of each number.

a. 30: _____ **b.** 28: _____

c. 45: _____ **d.** 17: _____

e. 48: _____ **f.** 42: _____

2. Find the GCF of each pair.

a. 12 and 20: ___ **b.** 18 and 27: ___ **c.** 16 and 40: ___

d. 14 and 25: ___ **e.** 36 and 48: ___ **f.** 30 and 45: ___

3. Write the prime factorization of each number. Use exponents.

a. 36 = _____ **b.** 50 = _____ **c.** 72 = _____

4. **Snack Bags.** Mr. Diaz has 36 granola bars and 54 juice boxes. He makes identical snack bags with nothing left over. What is the greatest number of bags he can make? What is in each bag?

Show your work.

Answer_____

5. **Find the Error.** Hana says the GCF of 15 and 28 is 0 because they share no factors. Explain her mistake and give the GCF.

_____

_____

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #___

AT HOME Ask someone at home to name two numbers. Find their GCF together.

Name_____ Date_____ Period_____

LESSON 6-7 · HOMEWORK PACKET · SHEET 2 OF 2

Multiples and Least Common Multiple

Maryland Standard 6.NS.B.4 · Learning target – I can list multiples of two numbers and find their least common multiple.

REMEMBER

- **Multiples** of a number come from skip-counting: 6, 12, 18, 24, …
- The **least common multiple** (LCM) is the least number that is a multiple of both.
- Use the LCM when events **happen together again**, or to buy packs for **equal amounts**.

WORKED EXAMPLE

Find the LCM of 8 and 12.

Step 1 Multiples of 12: 12, 24, 36, …

Step 2 12 is not a multiple of 8. 24 is (8 × 3 = 24).

Answer **LCM = 24**

PRACTICE

show your work · label units

1. Find the LCM of each pair.

a. 3 and 7: ___ **b.** 4 and 10: ___ **c.** 6 and 8: ___

d. 5 and 15: ___ **e.** 9 and 12: ___ **f.** 10 and 25: ___

2. **Cookout.** Hot dogs come in packs of 8. Buns come in packs of 10. Malik wants the same number of hot dogs and buns. What is the fewest packs of each he can buy?

Show your work.

Answer_____

3. **Blinking Lights.** One light blinks every 6 seconds. Another blinks every 9 seconds. They just blinked together. In how many seconds will they blink together again?

Answer_____ seconds

4. Two runners start together. One laps the track every 4 minutes, the other every 6 minutes. Which tells when they are next at the start together?

□ GCF of 4 and 6 □ LCM of 4 and 6

5. **Reason.** Is the LCM of two numbers always their product? Use 4 and 5, and 4 and 6, to explain.

_____

_____

TEST PRACTICE

choose the best answer

6. What is the LCM of 6 and 15?

A 3 **B** 30 **C** 90 **D** 21

7. Which pair of numbers has an LCM of 24?

A 4 and 6 **B** 6 and 8 **C** 2 and 12 **D** 24 and 48

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___